An Introduction to Ergodicity Economics

Ole Peters and Alexander Adamou

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Long-term growth

$\boxed{\text{Axiom: Agents maximize wealth.}}$

<u>Twist:</u> not expected wealth but real wealth over time.

Interesting because

Ergodicity is broken
$\rightarrow$ expected wealth $\neq$ real wealth over time.

Why the fuss?

"The economy is not ergodic."

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We investigate *standard mathematical models* from economics.
Finding: ergodicity is broken *there*.

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Ergodicity = property of mathematical models.

We investigate *standard mathematical models* from economics.
Finding: ergodicity is broken *there*.

→ Implication: standard analyses and results in economics are not valid. We re-do them.

Coin tosses

- The infamous coin toss
- The leveraged coin toss
- The cooperating coin toss
- The social coin toss

The infamous coin toss

Let us define our model of the dynamics of your wealth under the rules we just specified.[1] At regular intervals of duration δt, we randomly generate a factor r with each possible value $r_i \in \{0.6, 1.5\}$ occurring with probability $1/2$,

$$r = \begin{cases} 0.6 & \text{with probability } 1/2 \\ 1.5 & \text{with probability } 1/2 \end{cases} \tag{1.1}$$

and multiply current wealth by that factor, so that

$$x(t + \delta t) = x(t)r. \tag{1.2}$$

The infamous coin toss

Two properties:

- 1. Expected value
  $\langle x(t+1)\rangle = \frac{1}{2}(r_H x(t) + r_T x(t) = 1.05x(t)$.
  By induction, $\underline{\langle x(t)\rangle = x(0) \times 1.05^t}$.
  Per-round growth $g_{\langle\rangle} = 1.05$.

- 2. Time-average growth
  $x(t) = x(0) \times \prod_{\tau=1}^{t} r_\tau = x(0) \times r_H^{n_H} \times r_T^{n_T}$.
  per-round growth $g_t = \left(\frac{x(t)}{x(0)}\right)^{1/t} = r_H^{n_H/t} \times r_T^{n_T/t}$
  long-time limit
  $\underline{g = \lim_{t\to\infty} r_H^{n_H/t} \times r_T^{n_T/t} = r_H^{1/2} \times r_T^{1/2}} \approx 0.949$

The infamous coin toss

Comments:

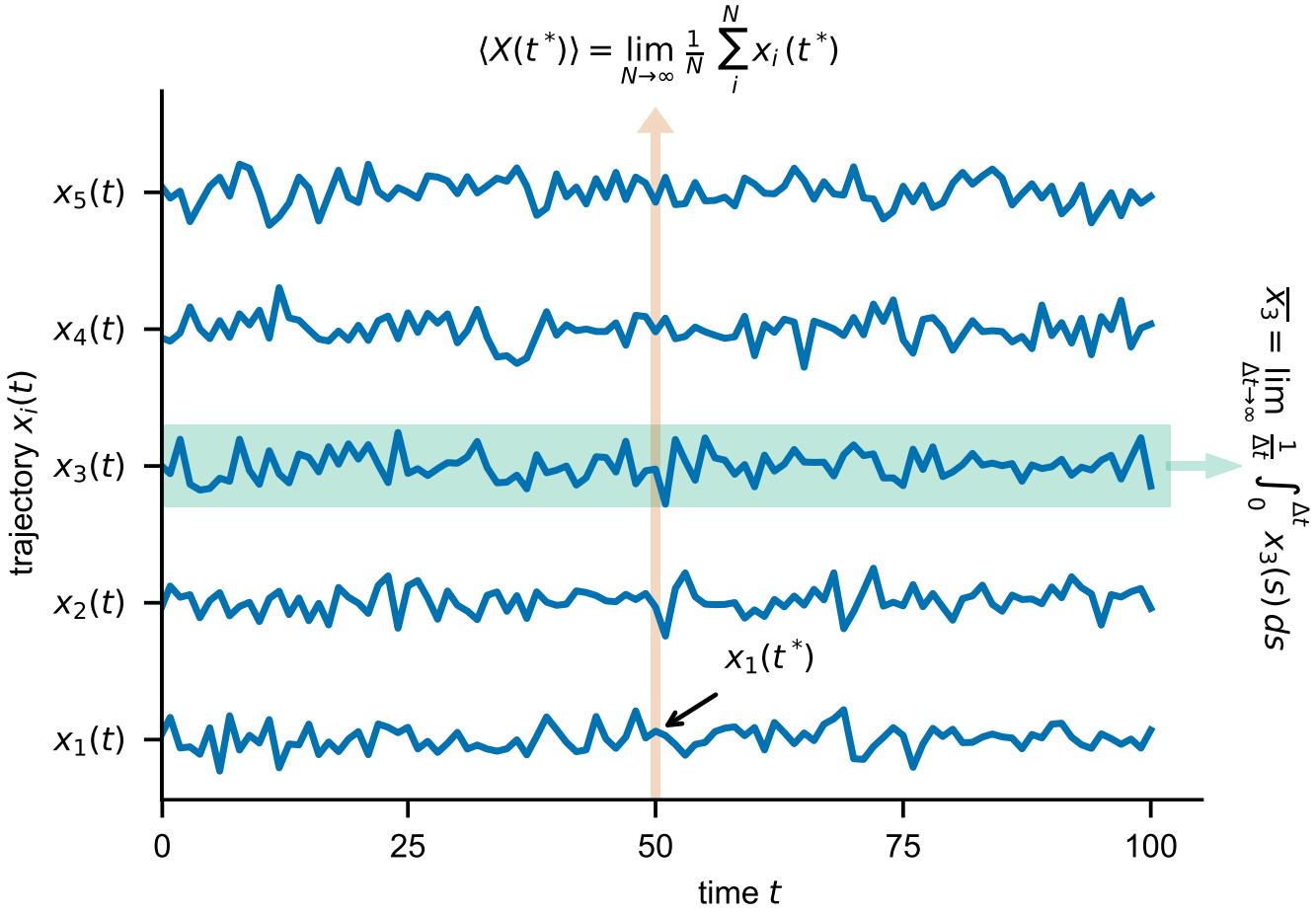
- discrete form of Geometric Brownian Motion (GBM): applies to all of finance theory.

The infamous coin toss

Comments:

- discrete form of Geometric Brownian Motion (GBM): applies to all of finance theory.
- multiplicativity: evolution, life, investing, learning, growing.

Averages


$\langle X(t^*)\rangle = \lim_{N\to\infty} \frac{1}{N} \sum_i^N x_i(t^*)$
$x_5(t)$
$x_4(t)$
$x_3(t)$
$x_2(t)$
$x_1(t)$
$x_1(t^*)$
trajectory $x_i(t)$
0
25
50
75
100
time t
$\overline{x_3} = \lim_{\Delta t\to\infty} \frac{1}{\Delta t} \int_0^{\Delta t} x_3(s)\, ds$

Chapter 3, p.25: Ergodicity

Richard H Thaler
@R_Thaler

This is NNT hogwash.

Philosophy: keep the text light.
Relevant ergodicity problem: expected value $\neq$ time average.
But there's 150 years of detailed theory behind this.

Dynamical systems

Dynamical sytem: (X, F, μ, T)
Mental image: $X =$ all possible identities x_i (you, me; this dice, that dice...). $F =$ coarse-graining: any identity is infinitely fine-grained and particular. Interesting statements can be made about collections $A \in X$ of identities (men, women; six-sided dice, dodecahedral dice...). μ size of a collection, $\mu(A)$ $T =$ time passing. Identities change in time as $T^n(A)$. These changes don't affect the size of the collection, $\mu(T^{-1}A) = \mu(A)$.
Mathematically, each entity marches through time deterministically. But that's not physically visible. Think of x_i as deeply hidden and unobservable, "the true you." But we can make measurements on x_i "how tall are you?"

Ergodic hierarchy

Ergodicity means

- Metric transitivity

Ergodic hierarchy

Ergodicity means

- Metric transitivity "Any differences are transitory."

  For any sets $A, B \in F$, there exists an n such that $\mu(T^{-n}(A) \cap B) > 0$ (mixing).

  Or, equivalently, if A satisfies $T^{-1}(A) = A$, then either $\mu(A) = 1$ or $\mu(A) = 0$ (only trivial invariant sets).

Ergodic hierarchy

Ergodicity means

- Metric transitivity

→ Equality of distributions

"In the long run, measurements in any two systems show the same values equally often."

Ergodic hierarchy

Ergodicity means

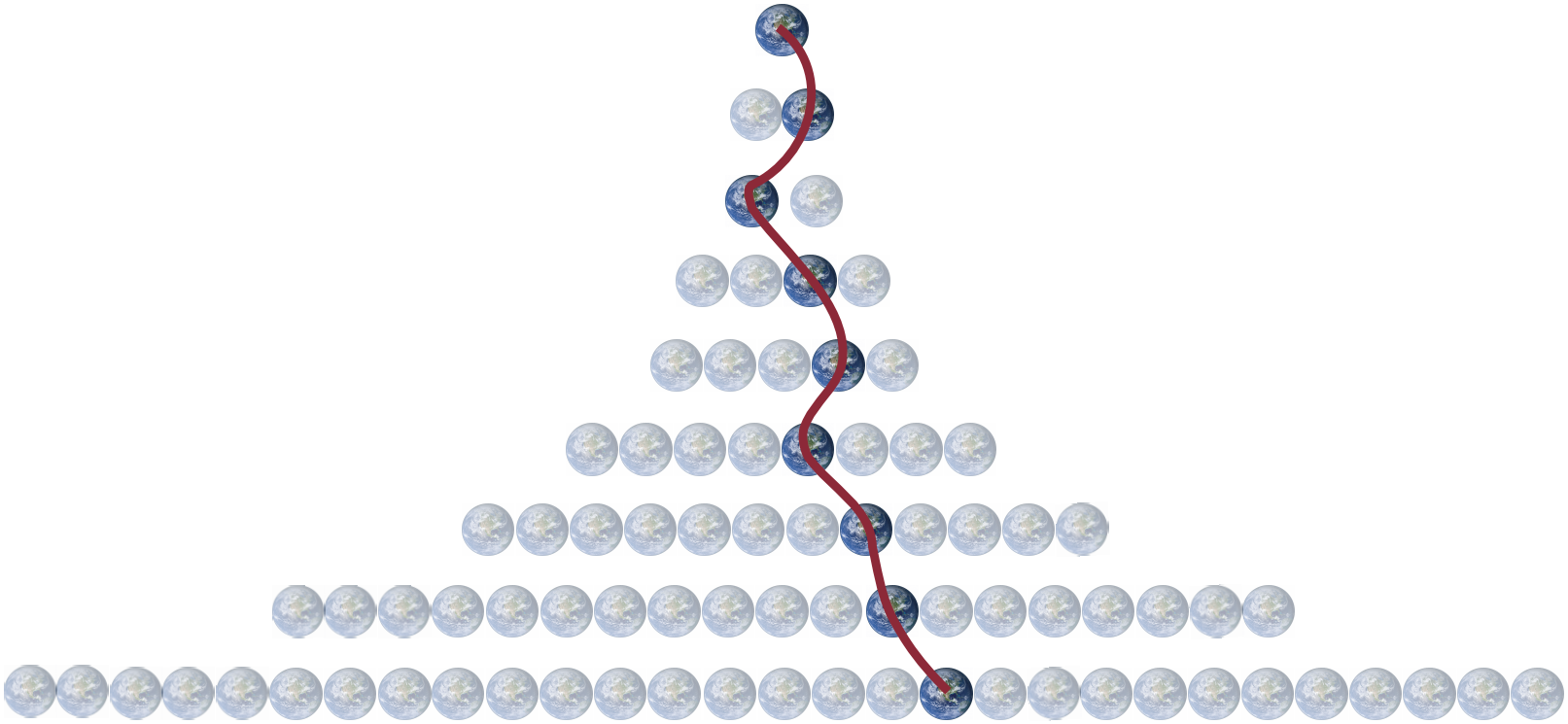
- Metric transitivity

→ Equality of distributions

→ Equality of averages "Averaging over time is equivalent to averaging over the ergodic ensemble."

Chapter 4, p.34: Models

More tools.

- Brownian motion
- Stochastic differential equations
- Itôcalculus
- Geometric Brownian motion

Thank you.